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Operating Water Distribution Systems for Equitable Access to Clean Water

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ABSTRACT

Urban water distribution systems (WDSs) are designed to deliver potable water to all end users. Unpredicted changes in water demands and hydraulics can increase residence time in pipes (water age), leading to growth of microbes and decreased water quality at some locations in a network. In response to reduced water quality, consumers may reduce demands for drinking, cooking, and cleaning. Lack of access to clean water can create high costs for some households due to the cost of using bottled water, paper plates, and laundromats for daily activities. This research develops an optimization framework to design operational strategies that maximize equity in a community that uses a WDS. Reduced demands and inequitable access to clean water are explored in this research in the context of the COVID-19 pandemic through a coupled framework. First, an agent-based modeling (ABM) framework is applied to simulate COVID-19 transmission, social distancing decision-making, and reductions in water demands. Large-scale reductions in demands, especially in industrial and commercial areas as individuals worked from home, leads to hot-spots of increased water age. The ABM is extended in this work to simulate households that choose to reduce demand from the system by buying bottled water for cooking, cleaning, and hygienic purposes. Equity is evaluated using an adjusted income metric that includes the cost of water bills and supplemental bottled water. A graph theory approach is applied to open and close valves to maximize equity. The coupled framework is applied for a virtual water distribution system, and results demonstrate operational strategies that improve equity for a community. This research develops an equity metric that assesses the water quality of delivered water and can be used to facilitate WDS management that provides equitable access to clean water.

1 INTRODUCTION

Water distribution systems (WDSs) are designed to deliver clean and safe water to all end users, yet water quality is susceptible to changes in system demand. Decisions made by end users to increase or decrease water use change the total demand required by the system and affect the flows in different locations in the network. Demand reductions cause increases in water age near

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the location of the demand change, leading to local water quality deterioration [1]–[4]. Previous work has shown that changing demands caused by COVID-19 social distancing resulted in increased water age, primarily in residential areas close to industrial sites [5], [6]. Low-income and minority households have historically been located near industrial sites, making increased water age at industrial-adjacent residential homes an equity issue. In addition, low-income and minority households also consume disproportionate amounts of bottled water [7]–[12]. In response to reduced water quality, consumers may increase bottled water use, which leads to increased costs to access water for the end user. The complex interactions between water quality, bottled water buying decisions, and the hydraulic network supplying water requires a sociotechnical framework to capture the personal decision making that is critical to the system.

The goal of this research is to develop a framework to assess the potentially inequitable water quality impacts of COVID-19 social distancing. Equity has previously been defined at access to high quality water for all end users, considering past and present contexts [13]–[15]. These contexts primarily include income and race differences that have historically affected access to water. Assessing equity is a primary goal of this research because past injustices such as the Flint water crisis have caused marginalized communities to decrease trust in WDS operation and increase risk perceptions of delivered water quality [16]. Including race and income within this modeling framework allows for optimization using equity as an objective function. Equity centric optimization leads to results that ensure high quality water to all end users while accounting for historical injustices with industrial facility siting.

The framework developed in this research builds on previous work that simulated COVID-19 transmission, social distancing decision-making, and hydraulic system operation using an agent-based model (ABM) [6]. We build on this work by adding a tap water avoidance model using Bayesian belief network (BBN) models. Water age is quantitatively used to model agent perceptions of water quality, and risk perceptions are coupled with water quality to predict tap water avoidance. An equity metric is developed using the adjusted income of each agent, and pressure reducing valves are strategically placed to reduce inequities [17], [18]. The new framework is applied for a case study to demonstrate disparities in access to safe and affordable water and the effect of strategic infrastructure operations on improvements to equity.

2 MATERIALS AND METHODS

An ABM framework is developed to model inequities in delivered water quality, and system optimization is performed using an equity-based objective function. The ABM framework builds on previous work that incorporated a susceptible-exposed-infected-recovered (SEIR) COVID-19 transmission model, a BBN-driven social distancing model, and a hydraulic model [6]. This work adds a cost of water framework that simulates tap water avoidance as a protective behavior when water quality is degraded due to COVID-19 social distancing dynamics (Figure 1). Water age is used to model water quality, and agents use perceptions of water quality and risk perceptions of tap water to make bottled water buying decisions. Households that buy bottled water use less tap water, further changing the hydraulics of the network including the water age. The cost of buying bottled water and tap water is modeled and compared to household income using an adjusted income metric. The adjusted income metric is used as an optimization objective to address inequities in water cost. New modules are described in the following sections, and parameters that are included in the modules are listed in Table 1.

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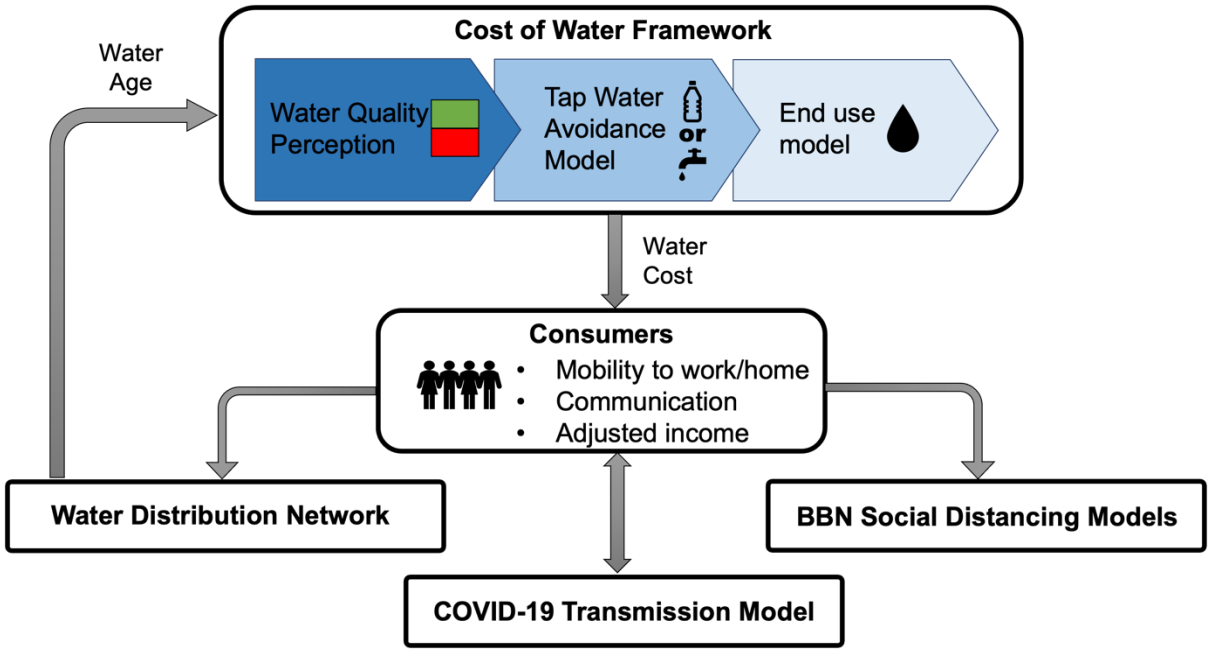

Figure 1: ABM framework integrates a new module to update the cost of water.

Table 1: ABM parameters are used to model tap water avoidance and adjusted income

Parameter	Symbol	Value
Water quality perception	P_{wq}	[good, poor]
Risk perception	P_r	[low, high]
Bottled water buying decision	D_{bw}	[buy, don't buy]
Water cost	C_W	Eq. 3
Adjusted income	I_A	Eq. 2

2.1 Cost of Water Framework

Household water cost is modeled using a framework that simulates agent perceptions of water quality, which informs agent decision-making on tap water avoidance behaviors and leads to changes in the cost of water. Agents perceive water quality using the water age at the node they occupy. Agents use the water quality perceptions and risk perceptions to decide to buy and use bottled water. Buying and using bottled water changes demand and household water expenditure.

2.1.1 Water Quality Perception

Each agent perceives water quality as a binary, good or poor quality. Water age directly changes the quality perception using a 180 hour threshold, shown in Eq. 1 [3]. When the water age (A_w) at the node an agent occupies exceeds these thresholds, the water quality perception (P_{wq}) changes.

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$$P_{wq} = \begin{cases} Good, & if\ A_w < 180\ hrs \\ Poor, & if\ A_w \geq 180\ hrs \end{cases} \quad (1)$$

2.1.2 Tap Water Avoidance Model

Bayesian belief network (BBN) models were used to calculate the probability that an agent buys bottled water, a tap water avoidance behavior. BBN models were trained with data from the 2015 American Housing Survey (AHS). Two predictors are modeled, risk perception and water quality perception, to predict bottled water buying behavior (Figure 2). Flood risk perception is used as a proxy for tap water risk perception as it was shown to correlate with water safety [9]. The mapping between BBN variable and AHS question is presented in Table 2.

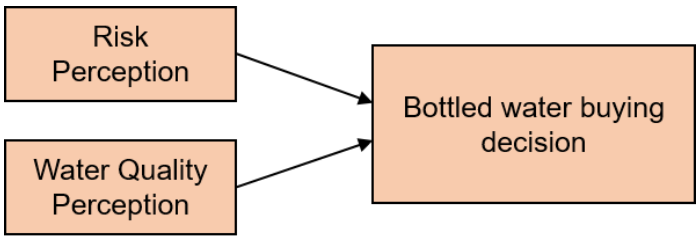

Figure 2: Bottled water buying decision-making BBN model.

Table 2: Responses to survey questions are used as variables in the BBN model shown in Figure 2.

BBN Model Variable	2015 AHS Question
Water Quality Perception	In your opinion, is the water from the tap safe for cooking and drinking? [Yes, no]
Risk Perception	This neighborhood is at high risk for floods or other disasters. [Agree, disagree]
Bottled water buying decision	Where do you get your water for drinking? [tap water, commercial bottled water]

Agents are assigned a risk perception value during model initialization. *Water quality perception* is a binary variable and is updated using Eq. 1. The bottled water buying decision is updated at each hourly time step based on the output probability of the BBN model (Figure 2).

2.1.3 End use Model

Agents that select to buy bottled water reduce demand on the hydraulic system and increase the cost of water for the household. The demand at nodes occupied by agents that have decided to buy bottled water is reduced by the minimum requirement for cooking and drinking, 15 L per person per day [19]. The reduction occurs at predefined hours to represent agent drinking and cooking habits.

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2.2 Equity Metric: Adjusted Income

Equity is assessed using an adjusted income metric. The adjusted income is given as:

$$I_A = I_G - C_W \tag{2}$$

where I_A is the adjusted income, I_G is the gross income and C_W is the cost of buying water. Each household is assigned an income value during model initialization. Households located closer to industrial sites have a higher chance of being assigned a lower income value to model historic siting inequities [20]. The cost of buying and using water is given as:

$$C_W = C_{TW} + C_{BW} \tag{3}$$

where C_{TW} and C_{BW} are the cost of purchasing tap water and bottled water, respectively. The cost used for bottled water is $0.325/L, an average established by the International Bottled Water Association [21]. The cost of tap water is modeled using an increasing block structure.

At the end of the simulation, the cost of water for each household is calculated and subtracted from the gross income to yield the adjusted income (Eq. 1). The adjusted income metric is used to analyze potential inequities caused by water age.

2.3 Valve Placement Optimization

A graph theory based algorithm is used to place pressure reducing valves (PRVs) throughout the system using adjusted income as an input [17]. PRVs are placed to reduce the pipe cost (see [17] and [18]), which includes the adjusted income. The disparities in adjusted income are reduced by strategically placed PRVs, which reduces the number of agents that adopt tap water avoidance behaviors.

3 EXAMPLE APPLICATION

The virtual city Micropolis is used as the model community for this research and includes the hydraulic network (Figure 3). Micropolis was developed as a realistic water network used for WDS security simulations. There are 458 terminal nodes (434 residential, 15 industrial and nine commercial) and the total daily demand is 4.53 ML/day. Micropolis represents a small city with a population of 4,606 agents modeled in the ABM.

4 PRELIMINARY RESULTS

Two scenarios were created to explore equity, a base case with no social distancing modeled and a prevention measure (PM) case with COVID-19 social distancing modeled. The PM case modeled working from home, dining out less, shopping for groceries less, and wearing PPE using BBN models for agent decision-making. The water age is reported for each node at the end of the simulation (Figure 3). Water age values at nodes in the box labeled (1) are greater than those outside box (1) in Figure 3b. The nodes in box (1) include many of the industrial nodes and the industrial-adjacent residential nodes and show inequities in the delivered water quality. Results that explore the impact of bottled water buying behaviors on water demands and equity will be reported in a conference presentation.

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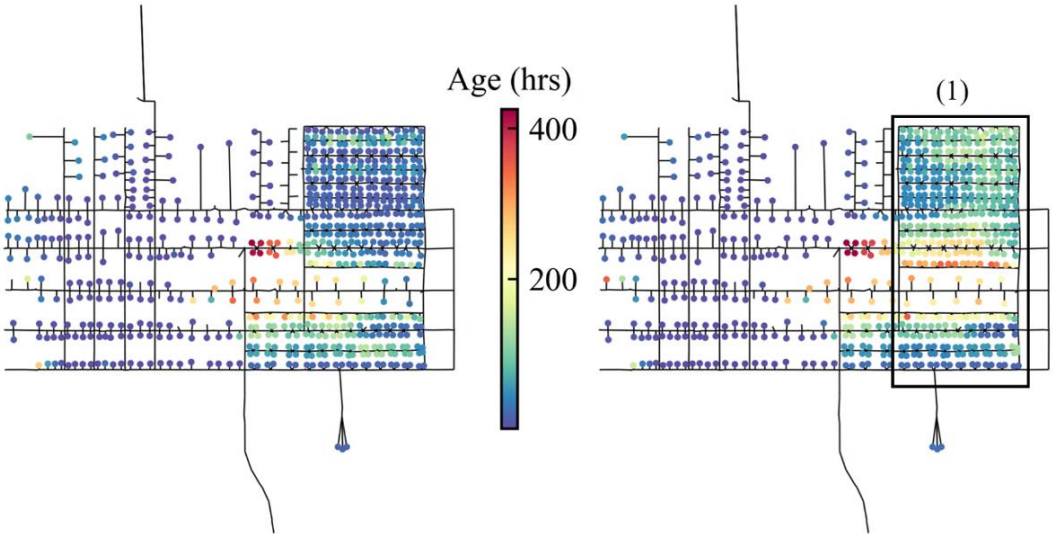

Figure 3: Virtual city of Micropolis showing the location of residential, commercial, and industrial nodes. Nodal water age for the final time step of the a) base case and b) PM case.

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